

Lecture 2

EX (BACK TO BINOMIAL) BINOMIAL PROB. MASS FUNK.

$$\Rightarrow f(x; \theta) = \binom{n}{x} \underbrace{(1-p)^m}_{\substack{C(p) \\ \text{ONLY DEPENDS ON PARAMETER}}} \exp \left\{ x \cdot \underbrace{\ln \left(\frac{p}{1-p} \right)}_{\eta} \right\}$$

THE NATURAL PARAMETER IS $\eta = \ln \left(\frac{p}{1-p} \right)$ (LOG-ODDS $\approx$ LOGIT)

SUCCESS
FAILURE
ODDS

$$\text{THEN } e^{\eta} = \exp \left\{ \ln \left(\frac{p}{1-p} \right) \right\} = \frac{p}{1-p} \stackrel{\text{ODDS}}{=}$$

$$\Leftrightarrow \frac{1}{p} - 1 = e^{-\eta}$$

$$\Leftrightarrow \frac{1}{p} = 1 + e^{-\eta}$$

$$\Leftrightarrow p = \frac{1}{1 + e^{-\eta}} = \frac{1}{1 + \frac{1}{e^{\eta}}} = \frac{e^{\eta}}{1 + e^{\eta}}$$

$$C(p) = (1-p)^m \Rightarrow \underbrace{C^*(\eta)}_{\text{"C-STAR"}} = \left(1 - \frac{e^{\eta}}{1 + e^{\eta}} \right)^m$$

INSTEAD OF P I WRITE THE OTHER FUNCTION

$$= \left(\frac{1 + e^{\eta} - e^{\eta}}{1 + e^{\eta}} \right)^m = \frac{(1 + e^{\eta})^{-m}}{(1 + e^{\eta})^{-m}}$$

C^* (C-STAR) FUNCTION IN TERMS OF η IS THE PREVIOUS $C(p)$ FUNCTION WITH p REPLACED WITH THE SAME VALUE IN TERMS OF η

THE NATURAL FORM OF BINOMIAL:

$$\Rightarrow f(x; \eta) = \binom{n}{x} (1 + e^{\eta})^{-m} \exp \{ x \cdot \eta \}$$

$\eta = \text{LOGIT} = \ln \left(\frac{p}{1-p} \right)$

APPLY RESULT OF LAST TIME:

$$E[t(x)] = E(x) = - \frac{\partial}{\partial \eta} \ln C^*(\eta)$$

$t(x)$ IS THE FUNCTION OF x THAT YOU SEE IN THE EXPONENTIAL FUNCTION
IN THIS CASE IS JUST x

$$= - \frac{\partial}{\partial \eta} \ln (1 + e^{\eta})^{-m}$$

$$= - \frac{\partial}{\partial \eta} (-m) \cdot \ln(1 + e^{\eta})$$

LIKE A CONST.

$$= m \cdot \frac{e^{\eta}}{1 + e^{\eta}} = mp$$

DERIVATE OF LOG

FROM BEFORE $p = \frac{e^{\eta}}{1 + e^{\eta}}$

EXPECTED VALUE OF x BINOMIAL RANDOM VARIABLE

EX (BACK TO NORMAL)

$$f(x; \mu, \sigma^2) = \underbrace{h(x)}_1 \cdot \underbrace{C(\mu, \sigma^2)}_{\text{BELOW (COMPLEX)}} \exp \left\{ \underbrace{w_1(\mu, \sigma^2)}_{\substack{\mu/\sigma^2 = \eta_1 \\ \text{FIRST NATURAL PARAMETER}}} \cdot \underbrace{t_1(x)}_x + \underbrace{w_2(\mu, \sigma^2)}_{\substack{1/\sigma^2 = \eta_2 \\ \text{SECOND NATURAL PARAMETER}}} \cdot \underbrace{t_2(x)}_{-x^2/2} \right\}$$

FROM $\frac{\mu}{\sigma^2} = \eta_1$ FROM $\frac{1}{\sigma^2} = \eta_2$

$$\begin{cases} \textcircled{1} \mu = \eta_1 \sigma^2 = \eta_1 \cdot \frac{1}{\eta_2} = \eta_1 / \eta_2 \\ \textcircled{2} \sigma^2 = 1 / \eta_2 \end{cases} \quad \left[\begin{array}{l} \text{WE WANT TO REWRITE THE FUNCTION} \\ C \text{ TO HAVE } C^* \end{array} \right]$$

FROM LAST LECTURE $\longrightarrow C(\mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left\{-\frac{1}{2} \frac{\mu^2}{\sigma^2}\right\}$

$$C^*(\eta_1, \eta_2) = \frac{\sqrt{\eta_2}}{\sqrt{2\pi}} \exp\left\{-\frac{1}{2} \frac{(\eta_1 / \eta_2)^2}{1 / \eta_2}\right\} = \frac{\sqrt{\eta_2}}{\sqrt{2\pi}} \exp\left\{-\frac{1}{2} \frac{\eta_1^2}{\eta_2}\right\}$$

C^* FUNCTION. ALLOW ME TO REWRITE THE NORMAL DENSITY IN ITS NATURAL FORM

NATURAL FORM OF NORMAL PDF:

NOW PARAMETRIZED ACCORDING TO η_1 AND η_2

$$f(x; \eta_1, \eta_2) = \frac{\sqrt{\eta_2}}{\sqrt{2\pi}} \exp\left\{-\frac{1}{2} \frac{\eta_1^2}{\eta_2}\right\} \cdot \exp\left\{\eta_1 x - \frac{\eta_2 x^2}{2}\right\}$$

$$E[t_1(x)] = E[X] = -\frac{\partial}{\partial \eta_1} \ln C^*(\eta_1, \eta_2)$$

WE HAVE 2 NATURAL PARAMETER

- $t_1(x) = x$
- $t_2(x) = x^2/2$

$$= -\frac{\partial}{\partial \eta_1} \ln \left[\frac{\sqrt{\eta_2}}{\sqrt{2\pi}} \exp\left\{-\frac{1}{2} \frac{\eta_1^2}{\eta_2}\right\} \right]$$

$$= -\frac{\partial}{\partial \eta_1} \left[\underbrace{\frac{1}{2} \ln \eta_2}_{\textcircled{I}} - \underbrace{\ln \sqrt{2\pi}}_{\textcircled{II}} - \underbrace{\frac{1}{2} \frac{\eta_1^2}{\eta_2}}_{\textcircled{III}} \right]$$

$$= \underbrace{0}_{\textcircled{I}} + \underbrace{0}_{\textcircled{II}} + \eta_1 / \eta_2 = \mu \quad \left[\begin{array}{l} \text{WE KNOW THAT THE EXPECTED} \\ \text{VALUE OF A NORMAL RANDOM} \\ \text{VARIABLE IS } \mu \end{array} \right]$$

$$E[t_2(x)] = E\left[-\frac{x^2}{2}\right] = -\frac{\partial}{\partial \eta_2} \left[\frac{1}{2} \ln \eta_2 - \ln \sqrt{2\pi} - \frac{1}{2} \frac{\eta_1^2}{\eta_2} \right]$$

$$= -\frac{1}{2\eta_2} - \frac{1}{2} \frac{\eta_1^2}{\eta_2^2} = -\frac{1}{2} \sigma^2 - \frac{1}{2} \mu^2$$

SECOND NON-CENTRAL MOMENT OF A NORMAL RANDOM VARIABLE

THEN $E(x^2) = -2 \cdot E\left[-\frac{x^2}{2}\right] = \sigma^2 + \mu^2$

SAMPLING

LINK WHAT WE HAVE DONE WITH THE ACTUAL DATASET

DATASET :



FEATURES WHAT YOU MEASURE

RANDOM VARIABLE $\sim$ EXP. FAMILY

WE COLLECT A RANDOM SAMPLE, A SUBSET OF THE POP.

WHERE UNITS CAN BE EXTRACTED $\begin{cases} \text{WITH REPLACEMENT} \\ \text{WITHOUT REPLACEMENT} \end{cases}$ (UNIT SELECTED TO BE PART OF THE SAMPLE, CAN BE SELECTED AGAIN)

IN BIG POPS : DISTRACTION IRRELEVANT $\rightarrow$ WE WILL ALWAYS DEAL WITH REPLACEMENT

$\Rightarrow$ A RANDOM SAMPLE OF SIZE m $\underline{X} = (X_1, X_2, \dots, X_m)$ $\rightarrow$ WE ARE ALWAYS TALKING ABOUT THE SAME RV. THESE ARE REPEATED MEASUREMENT OF THE RV.
EX: (AVERAGE GRADE STUD 1, ..., AVERAGE GRADE STUD. m)

$X_i, i=1, \dots, m$ ARE ASSURED TO BE i.i.d.
 $\downarrow$ $\downarrow$
INDEPENDENT IDENTICALLY DISTRIBUTED
THE VALUE OBSERVED ON ONE STUDENT WILL NOT AFFECT THE VALUE ON ANOTHER STUDENT (X_i DOES NOT AFFECT $X_j, j \neq i$)
(SAME f FOR ALL X_i)
 $\rightarrow$ IF I SELECT A SPECIFIC DISTRIBUTION FOR ONE STUDENT, THAT DISTRIBUTION IS VALID ALSO FOR ANOTHER ONE

NOTE: IN FINITE POPS, YOU NEED REPLACEMENT, OTHERWISE YOU DO NOT HAVE INDEPENDENT UNITS

WE DISTINGUISH BTW $\overset{\text{BIG}}{\underline{X}} = (X_1, X_2, \dots, X_m)$ RANDOM SAMPLE $\rightarrow$ RANDOM VECTOR. SOMETHING THAT IS RANDOM WITH ITS DISTRIBUTION
AND $\overset{\text{SMALL}}{\underline{x}} = (x_1, x_2, \dots, x_m)$ OBSERVED SAMPLE $\rightarrow$ OBSERVED VECTOR. JUST A VECTOR OF NUMBERS, NO RANDOMNESS
(REALIZED VALUE OF THE RANDOM SAMPLE)

WE EXPLOIT INDEPENDENCE ASSUMPTION TO RECOVER PDF/PMF OF $\underline{X}$: $f(\underline{x}) = \overset{\text{ACROSS THE SAME } f}{f(x_1)f(x_2)\dots f(x_m)} = \prod_{i=1}^m f(x_i)$
I.I.D.

WHEN X_i BELONGS TO THE EXPONENTIAL FAMILY, $\forall i$
THEN $(X_1, \dots, X_m)$ BELONG TO THE EXP. FAMILY

$$f(x; \theta) = h(x)c(\theta) \exp \left\{ \sum_{j=1}^k w_j(\theta) t_j(x) \right\}$$

$$f((x_1, \dots, x_m); \theta) = \prod_{i=1}^m f(x_i; \theta) = \underbrace{\left[\prod_{i=1}^m h(x_i) \right]}_{h((x_1, \dots, x_m))} \cdot \underbrace{c(\theta)^m}_{c(\theta)} \cdot \exp \left\{ \sum_{j=1}^k w_j(\theta) \underbrace{\sum_{i=1}^m t_j(x_i)}_{t_j((x_1, \dots, x_m))} \right\}$$

THE SAME

$\underbrace{\sum_{j=1}^k w_j(\theta) \sum_{i=1}^m t_j(x_i)}_{\sum_{i=1}^m \sum_{j=1}^k w_j(\theta) t_j(x_i)}$

$$E\{p\} = \sum_{i=1}^m \left[\sum_{j=1}^n u_j(\theta) \cdot t_j(x_i) \right]$$

Ex:

X	$f(x)$
0	0.1
1	0.3
2	0.5
3	0.1

PROB. VALUES ASSOCIATED TO EACH OF THIS VALUES

SET OF ALL POSSIBLE SAMPLES

THE SAMPLE SPACE $\mathcal{X}$ OF A SAMPLE OF SIZE $m=2$

CONTAINS 16 POSSIBLE SAMPLES (4×4)

POSSIBLE VALUES EXTRACTED FOR X_1

POSSIBLE VALUES X_2

X_1	X_2	$f(x_1)$	$f(x_2)$	$f((x_1, x_2)) = f(x_1)f(x_2)$
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0 0 0.1 0.1

0.01

PROB. OF OBSERVING $(X_1, X_2) = (0, 0)$

0 1 0.1 0.3

0.03

$P((X_1, X_2) = (0, 1))$

$\vdots$

(SEE NOTES FOR THE WHOLE TABLE)

THE JOINT PROB. ASSOCIATED TO THE SAMPLE $(0, 1)$ IS 0.03

Ex:

$X \sim \text{Pois}(\lambda)$, $\lambda = 3.5$ THEN $f(x; \lambda) = \frac{\lambda^x e^{-\lambda}}{x!}$, $x = 0, 1, 2, \dots$

$\text{supp}(X) = \mathbb{N}_0$

$$= \frac{(3.5)^x e^{-3.5}}{x!}$$

THEN $\text{supp}(X) = \text{supp}((X_1, \dots, X_m)) = \mathbb{N}_0^m$

WHEN $m=4$ AND I OBSERVE $X = (2, 4, 3, 3)$

$$\text{THEN } f(X; \lambda) = \prod_{i=1}^4 f(x_i; \lambda) = \prod_{i=1}^4 \frac{(3.5)^{x_i} e^{-3.5}}{x_i!}$$

$$= \left(\frac{(3.5)^2 e^{-3.5}}{2!} \right) \cdot \dots \cdot \left(\frac{(3.5)^3 e^{-3.5}}{3!} \right) = 0.00163$$

FOR X_1

SAMPLE STATISTICS

→ TOOL TO SUMMARIZE DATA

DATASETS CAN BE LONG LISTS OF NUMBERS
(NOT OF IMMEDIATE COMMUNICATION)

⇒ USE A SAMPLE STATISTIC TO SUMMARIZE THE DATA

A SAMPLE STATISTIC IS A FUNCTION OF THE RANDOM SAMPLE

$$Y = t(X) = t(X_1, X_2, \dots, X_m)$$

THE MOST COMMON STATISTICS ARE :

• SAMPLE MEAN $\bar{X} = \frac{1}{m} \sum_{i=1}^m X_i$ ← $\mu=1$

• SAMPLE VARIANCE : $S^2 = \frac{1}{m-1} \sum_{i=1}^m (X_i - \bar{X})^2$

• SAMPLE STANDARD DEVIATION : $S = \sqrt{S^2}$

$\mu=2$
(Almost)

• NON-CENTRAL MOMENT OF ORDER μ : $M_\mu = \frac{1}{m} \sum_{i=1}^m X_i^\mu$

• CENTRAL MOMENT OF ORDER μ : $\bar{M}_\mu = \frac{1}{m} \sum_{i=1}^m (X_i - \bar{X})^\mu$

• ORDER STATISTICS : $Y_1 \leq Y_2 \leq \dots \leq Y_m$
 $\underbrace{\quad}_{\min\{X_i\}} \quad \underbrace{\quad}_{\max\{X_i\}}$

• SAMPLE MEDIAN : $Me = \begin{cases} Y_{\frac{m+1}{2}} & , m \text{ ODD} \\ \frac{Y_{\frac{m}{2}} + Y_{\frac{m}{2}+1}}{2} & , m \text{ EVEN} \end{cases}$

EX: IF $m=5 \Rightarrow \overset{2 \text{ Ys BELOW}}{Y_1 \leq Y_2} \leq \boxed{Y_3} \leq \overset{2 \text{ Ys ABOVE}}{Y_4 \leq Y_5}$

$$Me = Y_{\frac{5+1}{2}} = Y_3$$

EX: IF $m=4 \Rightarrow \overset{1 \text{ BELOW}}{Y_1} \leq \overset{1 \text{ ABOVE}}{\boxed{Y_2 \leq Y_3}} \leq Y_4$

$$Me = (Y_{\frac{4}{2}} + Y_{\frac{4}{2}+1}) / 2 = \frac{Y_2 + Y_3}{2}$$

SPECIAL CARE WILL BE DEDICATED TO $\bar{X}$ AND S^2

CARE: SAMPLE STATISTICS ARE RANDOM VARIABLES

$\left(\frac{1}{m} \sum_{i=1}^m (X_i - \mu)^2 \right)$ NOT A SAMPLE STATISTIC
SINCE IT DEPENDS ON
UNKNOWN μ

$\Rightarrow$ I don't know its
VALUE AFTER OBSERVATION
OF $\underline{X}$

THE DISTRIBUTION OF A SAMPLE STATISTIC
IS CALLED SAMPLING DISTRIBUTION:

FOR $Y = t(X)$, WE HAVE

$$F_Y(y) = P(Y \leq y) = P(t(X) \leq y) \\ = P(\underline{X} \in S_y), \quad S_y = \{x : t(x) \leq y\}$$

OFTEN: NOT POSSIBLE TO DERIVE IT, BUT

SOME EXCEPTIONS -

• FOR $\bar{X}$: $E(\bar{X}) = \mu$ (MEAN OF $\bar{x}$) (REGARDLESS OF THE DISTRIB. OF X)

SINCE $E(\bar{X}) = E\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n} \sum_{i=1}^n E(X_i) = \frac{1}{n} \sum_{i=1}^n \mu = \mu$

IDENTICALLY
DISTRIBUTED

ALSO: $V(\bar{X}) = \frac{\sigma^2}{n}$ (VARIANCE OF $\bar{x}$)

POPULATION VARIANCE σ^2 (MORE DATA $\rightarrow$ LESS VARIANCE FOR $\bar{X}$)

SAMPLE SIZE n

SINCE $V(\bar{X}) = V\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} V\left(\sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n V(X_i)$

INDEPENDENCE

IDENT. DISTRIBUTION $\Rightarrow \frac{1}{n^2} \sum_{i=1}^n \sigma^2 = \frac{\sigma^2}{n}$

• FOR S^2 : $E(S^2) = \sigma^2$ (SAMPLE VARIANCE)

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i^2 + \bar{X}^2 - 2\bar{X}X_i)$$

$$\begin{aligned}
&= \frac{1}{n-1} \left(\sum_{i=1}^n X_i^2 + \sum_{i=1}^n \bar{x}^2 - \sum_{i=1}^n 2\bar{x}X_i \right) \\
&= \frac{1}{n-1} \left(\sum_{i=1}^n X_i^2 + n\bar{x}^2 - 2\bar{x} \sum_{i=1}^n X_i \right) \quad \begin{array}{l} \text{from } \bar{x} = \sum_{i=1}^n X_i / n, \\ \sum_{i=1}^n X_i = n\bar{x} \end{array} \\
&\quad \quad \quad - 2n\bar{x}^2
\end{aligned}$$

$$= \frac{1}{n-1} \left(\sum_{i=1}^n X_i^2 - n\bar{x}^2 \right)$$

ALTERNATIVE FORMULA
FOR S^2

$$\begin{aligned}
\Rightarrow E(S^2) &= \frac{1}{n-1} \left(\sum_{i=1}^n E(X_i^2) - n E(\bar{x}^2) \right) \quad \begin{array}{l} V(\bar{x}) + E(\bar{x})^2 = \frac{\sigma^2}{n} + \mu^2 \\ \text{from above} \end{array} \\
&\quad \quad \quad \downarrow \text{FROM } V(X) = E(X^2) - E(X)^2 \Rightarrow E(X^2) = V(X) + E(X)^2 \\
&\quad \quad \quad \Rightarrow E(X_i^2) = \sigma^2 + \mu^2 \\
&= \frac{1}{n-1} \left(\sum_{i=1}^n (\sigma^2 + \mu^2) - n \left(\frac{\sigma^2}{n} + \mu^2 \right) \right) \\
&= \frac{1}{n-1} \left(n(\sigma^2 + \mu^2) - \sigma^2 - n\mu^2 \right) \\
&= \frac{1}{n-1} \cdot (n-1)\sigma^2 = \sigma^2
\end{aligned}$$